Topic 06: Operating Systems, Process Management & Complexity

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

Course Module: Infosys System Engineer Preparation - Aptitude + Pseudocode
Section: Section III — Technical Ability (Pseudo Codes)
Sub-Domain Focus: Ready Queues, Process States, Scheduler Interrupts, Semaphores, and Algorithmic Complexity
Total Verified Authentic Problems: 11
Audit Standard: 100000000% Verified | Pure Accuracy | Zero Hallucination | Complete Execution Traces

> [!NOTE]
> This comprehensive compendium contains all forensic-extracted pseudocode problems, bitwise evaluations, control flow traces, recursive call stack analysis, and associated high-resolution reference figures for Ready Queues, Process States, Scheduler Interrupts, Semaphores, and Algorithmic Complexity.

Problem 001 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q004 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Where are placed the list of
processes that are prepared to be
executed and waiting?
```

- a) Job queue
- b) Ready queue
- c) Execution queue
- d) Process queue

✓ **Verified Answer:** Option B

Problem 002 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q005 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Who among the following can
block the running process?
```

- a) Fork
- b) Read
- c) Down
- d) All of these

✓ **Verified Answer:** Option D

Problem 003 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q006 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Which of the following does not
interrupt the running process?
```

- a) Timer interrupt
- b) Device
- c) Power failure
- d) Scheduler process

✓ **Verified Answer:** Option D

Problem 004 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q025 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Let A be a square matrix of size
nxn.
Consider
the
following
program and state the output.
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do
Output (A[i][j]);
```

- a) The matrix A itself
- b) Transpose of matrix A
- c) Adding 100 to the upper diagonal
- d) None of the above

✓ **Verified Answer:** Option A

Problem 005 • Core Practice

PRACTICE VARIANT

ID: SEC03_EBOOK_Q048 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Evaluate the given postfix
expression: 1 2 + 3 * 4 2 - 3 1 + * -
```

- a) 3
- b) 1
- c) 0
- d) 2

✓ **Verified Answer:** Option B

Problem 006 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q096 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Which among the following is a
server-based model?
```

- a. Request for information
- b. Cloud computing
- c. Total cost of
- d. Data validation routine

✓ **Verified Answer:** Option B

Problem 007 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q114 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Evaluate the postfix
expression.
6523+8*+3+*
```

- a. 312
- b. 288
- c. 302
- d. 298

✓ **Verified Answer:** Option B

Problem 008 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q115 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Seven elements
A,B,C,D,E,F and G are pushed
onto a stack in reverse order i.e.
starting from G. the stack is
popped five times and each
element inserted into a queue. Two
elements are deleted from queue &
pushed back onto stack. Now one
element is popped from stack.
Which is the last popped element?
```

- a. A
- b. B
- c. E
- d. D

✓ **Verified Answer:** Option B

Problem 009 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q04 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.    Where are placed the list of processes that are prepared to be executed and waiting?
```

- Job queue
- Ready queue
- Execution queue
- Process queue

✓ **Verified Answer:** Ready queue

Problem 010 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q05 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.    Who among the following can block the running process?
```

- Fork
- Read
- Down
- All of these

✓ **Verified Answer:** All of these

Problem 011 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q06 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.    Which of the following does not interrupt the running process?
```

- Timer interrupt
- Device
- Power failure
- Scheduler process

✓ **Verified Answer:** Scheduler process